Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/13/2005
PI Name:	Gibson, Ursula Ph.D.		
Project Title:	In-situ Optical Waveguides for Promoting and Monitoring Protein Crystal Growth		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Hanover	**State:**	NH
Zip Code:	03755-8000	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	97-HEDS-02
Start Date:	03/01/1999	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	3
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1590		
Performance Goal No.:	1.21		
Performance Goal Text:	Sustain a leading biotechnology research program that will assure continued scientific and technical leadership.		
Task Description:	In this project we are using electromagnetic fields to test whether electrophoretic and dielectrophoretic effects can be used to enhance or lacalize protein nucleation by altering the local precipitant and protein concentration. At low frequencies, we are using surface plasmons to measure the changes of the dielectric constant of the solutions in the presence of an electric field. At higher frequencies, we use photolithographically defined electrodes, and at the upper limit, we use gradient optical fields to trap protein crystals and grow them in the trap. The objective of this work is to investigate the use of electric fields to influence the nucleation and growth of protein crystals. We explore a wide range of frequencies from dc to optical fields. If electric fields can be used to localize the nucleation sites for crystallization, experiments such as x-ray diffraction can		

	be carried out on crystals during growth. This will lead to more efficient use of space station equipment, and may advance ground-based protein crystallization techniques. Variation of pH in a single sample may also be a useful screening technique.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improved protein crystallization techniques are critical to the development of a variety of new pharmaceuticals.
Task Progress:	We have demonstrated the movement of proteins using direct current (dc) and high frequency electromagnetic fields. With dc fields, we were able to control a local concentration of the protein in solution. With proper design of electrodes, this technique might be developed to fix the location of nucleation of protein crystals, for in-situ analysis. We performed modeling on the motion of proteins and precipitant under the influence of dc and radiofrequency (RF) fields, and predict that these fields should be useful for concentrating protein in one location to control the nucleation of crystals. Preliminary measurements were made of the RF dielectric properties of lysozyme. We demonstrated growth of protein crystals in an optical trap, using the gradient in the electric field strength to hold a seed crystal in position. An increase of the supersaturation ratio led to crystal growth. Using the birefringence of the crystal, the orientation of the seed could be controlled during the growth process. This non-contact method hold the promise for detailed in-situ analysis of the growth of protein crystals.
Bibliography Type:	Description: (Last Updated: 10/29/2019)
Articles in Peer-reviewed Journals	Singer W, Rubinsztein-Dunlop H, Gibson U. "Manipulation and growth of birefringent protein crystals in optical tweezers." Opt Express. 2004 Dec 27;12(26):6440-5. https://doi.org/10.1364/OPEX.12.006440 ; PubMed PMID: 19488294 , Dec-2004